

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. EMERGENCY MEDICINE DELIVERY BY AI DRONE

1.1 Search Formulation

Element	Definition in the Drone Scenario
State	Drone position, remaining battery, current weather, known blocked paths and medicine status.
Initial state	Emergency base or temporary medical camp.
Goal state	Target relief location reached with medicine intact and minimum unacceptable risk.
Actions	Fly to a neighbouring waypoint, change altitude, wait briefly, or reroute.
Step cost	Travel time + energy consumption + terrain/weather risk penalty.
Heuristic $h(n)$	Estimated remaining cost, initially based on straight-line distance and improved using wind/risk information.
Performance measure	Fast, safe, reliable delivery with low total cost and minimal replanning overhead.

1.2 Greedy Best-First Search Behaviour

Priority rule	Choose node with minimum $h(n)$. Greedy Best-First Search expands the location that appears closest to the destination, without considering the cost already spent.
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- In this scenario, Greedy Best-First Search repeatedly selects the waypoint with the smallest straight-line distance to the medical destination. It may quickly fly toward the goal, but it ignores the accumulated cost $g(n)$.

- A route that appears geographically close may pass through strong wind, a flooded corridor or a high-risk area.
- When a blocked path is reported, the affected node must be rejected and the frontier reconsidered.

Strength	Why it helps under urgency
Fast directional progress	The heuristic directly pulls the search toward the destination.
Often expands fewer nodes	Useful when the map is large and a rapid approximate route is acceptable.
Simple implementation	Low decision overhead suits limited onboard computation.

Weakness	Practical consequence
Not optimal	May select a longer or more dangerous route because $g(n)$ is ignored.
Can be misled by obstacles	A straight-line heuristic may repeatedly approach a blocked region.
Incomplete in some unbounded spaces	Could keep following attractive alternatives without reaching the safest goal.
Uncertainty-sensitive	Partial updates can invalidate its current direction abruptly.

1.3 A* Search Behaviour

Evaluation	$f(n) = g(n) + h(n)$ $g(n)$ is the actual cost already incurred; $h(n)$ is the estimated remaining cost. A* balances experience so far with future promise.
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- A* compares routes using both accumulated and estimated remaining cost.
- For example, Route A may be 2 km closer to the goal but pass through heavy rain, while Route B is slightly longer but safer and less energy-intensive.

- When weather and risk are included in $g(n)$, A^* can prefer Route B even though its straight-line distance is larger. If $h(n)$ is admissible and consistent, A^* can return an optimal route for the currently known map. Risk-aware cost innovation: Define each edge cost as $C = w_1(\text{time}) + w_2(\text{energy}) + w_3(\text{weather risk}) + w_4(\text{terrain risk}) + w_5(\text{battery reserve penalty})$. This converts safety from a vague preference into a measurable search objective.

1.4 Impact of Heuristic Accuracy

Heuristic condition	Effect on Greedy Search	Effect on A^*
Highly accurate	Very fast and often near-optimal.	Expands fewer nodes while retaining optimality if admissible.
Underestimates remaining cost	Still may choose a poor route because $g(n)$ is ignored.	Safe for optimality, but may expand more nodes.
Overestimates remaining cost	Can become aggressively biased and unsafe.	May lose optimality, behaving more like Greedy Search.
Ignores wind/terrain	May head toward a deceptively close hazard.	$g(n)$ partly corrects the error, but performance decreases.
Updated in real time	Can redirect quickly but may oscillate.	Supports better replanning when combined with incremental search.

1.5 Dynamic Blocked Paths and Real-Time Updates

- A newly blocked path invalidates part of the current search tree. Restarting ordinary A^* from scratch is correct but computationally wasteful.
- A stronger operational design is to use incremental replanning, such as D^* Lite or Lifelong Planning A^* , which reuses previous search information and repairs only the affected region.

- The drone should also reserve battery for diversion and maintain a safe loiter or emergency-landing option.
1. Receive a blocked-path or weather update.
 2. Increase the cost of the affected edge or mark it unavailable.
 3. Invalidate route segments that depend on that edge.
 4. Replan from the current drone position using updated costs.
 5. Check battery reserve and risk threshold before executing the new route.

1.6 Recommendation and Final Decision

- Recommended algorithm: Risk-aware A* with continuous/incremental replanning.
- It gives the best balance of optimality, safety and real-time response. Greedy Best-First Search may be retained only as an emergency fallback when a result is needed within an extremely small time budget.
- The recommendation is justified because emergency medicine delivery cannot sacrifice safety merely for computational speed.
- A* explicitly accounts for already-incurred cost and can optimize a composite objective that includes delay, energy and risk. To meet real-time constraints, the system should use bounded computation, frequent sensor updates, incremental replanning and a weighted-A* mode when a slightly suboptimal but faster route is acceptable.

Criterion	Greedy Best-First	A* / Dynamic A*
Decision speed	Very high	High with incremental replanning
Optimality	Not guaranteed	Guaranteed for known map with admissible heuristic
Safety integration	Weak unless embedded in heuristic	Strong through path cost and constraints
Response to changes	May re-search inefficiently	Can repair route efficiently
Final suitability	Fallback only	Primary choice

2. SMART-CITY TRAFFIC SIGNAL OPTIMIZATION

2.1 Optimization Formulation

Decision vector	$X = \{\text{green time, red time, cycle length, phase order, offsets}\}$ for every intersection. A complete traffic plan is one candidate solution in the search space.
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Objective	Minimize $J(X) = \alpha W + \beta F + \gamma C + \delta Q + \varepsilon P$ W: waiting time, F: fuel use, C: congestion, Q: queue spillback, P: pedestrian/safety penalty. Weights express city priorities.
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- Constraints include minimum pedestrian crossing time, maximum cycle length, non-conflicting movements, emergency-vehicle priority, fairness across directions and hardware limits.
- Traditional systematic search is inefficient because every intersection has many possible timings; multiplying these choices across hundreds of intersections causes combinatorial explosion.
- Further, traffic demand changes while the algorithm is still searching, so a perfect solution for an old pattern may be obsolete.

2.2 Hill Climbing in the Scenario

- Hill Climbing begins with a feasible signal plan, changes one or more timings slightly, evaluates the new plan using the objective $J(X)$, and accepts a neighbour only if it improves performance.
- For example, it may add 5 seconds of green time to the east-west phase at a congested junction and retain that change if total delay decreases.

1. Generate an initial feasible traffic plan.
2. Evaluate total waiting time, fuel use and congestion.
3. Create neighbouring plans by modifying phase times or offsets.
4. Move to the best improving neighbour.
5. Stop when no immediate neighbour improves the objective.

Advantage	Limitation
Simple and computationally inexpensive	Can stop at a local optimum.
Produces quick incremental improvement	Does not naturally explore temporarily worse solutions.
Suitable for online fine-tuning	Sensitive to initial signal plan.
Easy to explain to operators	May oscillate under noisy traffic measurements.

2.3 Local Maxima, Plateaus and Ridges: Real-World Meaning

Search difficulty	Traffic interpretation	Effect
Local optimum	A timing plan works well for nearby junctions but causes congestion farther downstream.	Hill Climbing stops too early.
Plateau	Many small timing changes produce nearly identical waiting time.	No clear direction; progress becomes slow or random.
Ridge	Improvement requires coordinated changes at several intersections, but changing only one appears worse.	Single-step neighbourhood cannot reach the better region.
Dynamic objective	A sports event or accident changes traffic suddenly.	The earlier optimum becomes invalid.

2.4 Simulated Annealing and Genetic Algorithms

- Simulated Annealing sometimes accepts a worse solution with probability $P = \exp(-\Delta E/T)$. At high temperature T , it explores broadly; as T decreases, it becomes selective. This allows escape from local optima. However, tuning the cooling schedule for a large city is difficult and one candidate solution is improved at a time.

- A Genetic Algorithm maintains a population of signal plans. Selection keeps high-quality plans, crossover combines useful timing patterns from different plans, and mutation introduces diversity.
- Parallel evaluation makes it suitable for large networks. It can also preserve several strong alternatives for different traffic regimes rather than relying on one trajectory.

Feature	Simulated Annealing	Genetic Algorithm
Search style	Single-solution stochastic search	Population-based global search
Escapes local optima	By accepting worse moves	By diversity, crossover and mutation
Parallelism	Limited	Excellent
Adaptability	Good with reheating/restart	High through continuous evolution
Interpretability	Moderate	Moderate; elite solutions can be inspected
Large-network suitability	Good	Very good

2.5 Recommended Scalable Architecture

- Recommended approach: Adaptive Genetic Algorithm + local Hill-Climbing refinement (memetic approach).
 - The GA finds globally promising regions; Hill Climbing rapidly fine-tunes each elite solution.
1. Use live traffic sensors to estimate queues and arrival rates.
 2. Divide the city into interacting zones to reduce dimensionality.
 3. Evolve timing plans using a Genetic Algorithm in parallel.
 4. Apply local search to the best candidates for rapid refinement.
 5. Enforce hard safety constraints before deployment.
 6. Re-evaluate and evolve again when traffic distribution changes.

- This hybrid strategy is preferred because it combines scalability, global exploration and rapid local adaptation.
- A rolling-horizon controller can optimize the next 10-15 minutes using the newest data.
- Elite plans should be retained as “memory” for recurring conditions such as school opening, office peak hours, rain or stadium events.

3. AUTONOMOUS MARS ROVER IN UNKNOWN TERRAIN

3.1 Why Online Search Is Required

- Offline search assumes that the state space, actions and outcomes are sufficiently known before execution.
- That assumption fails on Mars. The rover sees only a limited area, communication delay prevents constant human control, wheel slip changes movement outcomes and new craters or rocks may appear after a planned path is generated.
- Therefore, the rover must perform online search: choose an action, observe the result, update its model and plan again.

Offline Search	Online Search
Plans before acting	Interleaves planning and action
Requires a known model or map	Learns unknown states and action outcomes
A route may become invalid before execution	Replans after each meaningful observation
Lower execution flexibility	High adaptability to hazards
Suitable for static, well-mapped worlds	Suitable for unknown and partially observable worlds

3.2 Challenges of Partial Observability and Dynamic Conditions

- Limited field of view: hazards beyond the sensing range are unknown.
- Sensor uncertainty: shadows, dust and lighting may cause false obstacle detection.
- Localization error: wheel slip can make the estimated position differ from the true position.
- Action uncertainty: the same motor command may produce different displacement on sand and rock.
- Resource limits: energy, computation, memory and communication bandwidth are restricted.
- Safety-critical exploration: one poor action may permanently trap or damage the rover.
- Environmental change: dust, slope stability and temperature can alter traversability.

3.3 Building and Updating the Internal Model

- The rover maintains a belief-based internal model rather than assuming one perfectly known state.
- Sensor readings from cameras, lidar/radar, inertial measurement and wheel odometry are fused to estimate position and terrain.
- Each observed cell is labelled with traversability, slope, hazard probability, scientific value and confidence.

Model component	Update performed
Occupancy / terrain map	Mark free, blocked and uncertain cells from sensors.
Localization belief	Update position probability using motion and observations.
Action-cost model	Learn energy use, slip and time for different terrain types.
Hazard map	Increase risk cost for steep slopes, loose soil and crater edges.
Scientific utility map	Assign higher reward to promising sample locations.
Experience memory	Store failed paths and successful traversals to improve future estimates.

1. Perceive the local environment.
2. Estimate the current belief state.
3. Update terrain and hazard probabilities.
4. Plan toward a safe subgoal within the known region.
5. Execute one or a few actions.
6. Compare expected and actual outcomes; learn and replan.

3.4 Comparison with Uninformed and Informed Search

Strategy	Usefulness	Limitation for Mars
Breadth-First Search	Complete for equal step costs	Memory-intensive and ignores terrain risk.
Uniform-Cost Search	Finds least-cost route in a known weighted map	Must expand many nodes and assumes stable known costs.
Depth-First Search	Low memory	Can enter dangerous dead ends; not optimal.
A* Search	Uses cost and heuristic efficiently	Needs a reasonably known map; must be repeatedly updated.
Online heuristic search	Acts, observes, learns and replans	May incur exploration cost but is suited to uncertainty.
Real-Time A* / LRTA*	Makes bounded-time decisions and learns heuristic values	Initial exploration can be suboptimal.

3.5 Recommended Safe Online Strategy

- Recommended approach: Model-based online heuristic search with risk-sensitive A*/D* Lite for local replanning, frontier exploration for map growth and a safety shield that rejects actions exceeding hazard thresholds.
- The rover should not always pursue the shortest path. A utility function should balance progress, energy, scientific value, communication opportunity and survival probability.
- Unknown terrain should carry a moderate exploration penalty, while high-risk terrain should be treated as a hard constraint unless no alternative exists.
- The rover should move toward safe intermediate subgoals and preserve a return path and energy reserve.

4. UNIVERSITY EXAMINATION TIMETABLE AS A CSP

4.1 CSP Formulation

CSP model	$CSP = (X, D, C)$ X is the set of variables, D the domains, and C the constraints.
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CSP element	Definition
Variables	One variable E_i for each examination/course. A compound variable may represent (date, session, hall).
Domains	Available date-session-hall combinations that satisfy basic capacity and facility requirements.
Hard constraints	No student overlap, no faculty conflict, hall capacity, hall availability, precedence and accommodation requirements.
Soft constraints	Spacing between exams, preferred sessions, balanced hall use, reduced consecutive exams and faculty preferences.
Goal	Assign every examination a valid domain value while minimizing soft-constraint penalties.

Constraint	Formal / practical meaning
Student conflict	For any courses i and j sharing a student: $Time(E_i) \neq Time(E_j)$.
Hall capacity	Assigned hall capacity $\geq$ number of registered students.
Hall exclusivity	Two exams cannot use the same hall in the same session unless partitioning is allowed.

Faculty availability	Exam session must belong to the invigilator/faculty available set.
Precedence	If subject A must be before B: $\text{Time}(A) < \text{Time}(B)$.
Special accommodation	Accessible hall, extra-time slot or isolated room must be available.
Spacing preference	Avoid assigning a student two exams on consecutive sessions where possible.

4.2 Backtracking Search

- Backtracking assigns one examination at a time. If a partial assignment violates a constraint, the algorithm reverses the most recent decision and tries another value. It is systematic and complete, but plain backtracking may be slow because it can discover conflicts late.
 1. Select an unassigned exam variable.
 2. Choose a candidate date-session-hall value.
 3. Check consistency with all currently assigned exams.
 4. If consistent, continue recursively.
 5. If no value works, undo the previous assignment and try another branch.
 6. Return a complete timetable when all variables are assigned.

4.3 Constraint Propagation and Forward Checking

- Forward checking looks ahead after every assignment. When Exam A is placed in Monday morning, that time is immediately removed from the domains of all exams sharing students or faculty with A.
- If any unassigned exam loses every legal value, failure is detected early and the algorithm backtracks immediately.
- Arc consistency (for example, AC-3) goes further by repeatedly removing values that have no supporting value in connected variables.
- Efficiency insight: Backtracking asks “Is the current choice legal?” Forward checking additionally asks “Will the remaining exams still have at least one possible choice?” This early warning prevents large useless subtrees.

4.4 Heuristics for High Scalability

Technique	Role in the timetable
MRV - Minimum Remaining Values	Schedule the most constrained exam first, such as a large course with few suitable halls.
Degree heuristic	When tied, choose the exam that conflicts with the largest number of others.
LCV - Least Constraining Value	Choose the slot that removes the fewest choices from neighbouring exams.
Forward checking	Prune inconsistent domain values immediately.
Arc consistency	Propagate pairwise constraints more deeply.
Conflict-directed backjumping	Jump back to the variable responsible for failure instead of one level only.
Min-conflicts local search	Efficiently repair a nearly complete large timetable.
Problem decomposition	Split weakly connected departments, then coordinate shared courses and halls.

4.5 Real-World Challenges and Best Strategy

1. Late registration changes alter conflict sets and hall requirements.
 2. A faculty member may become unavailable after timetable publication.
 3. Soft constraints may conflict; perfect satisfaction may be impossible.
 4. Large common courses create high-degree nodes in the conflict graph.
 5. Special accommodations require fairness and privacy-aware handling.
 6. Changes should minimize disruption to already communicated exams.
- Recommended strategy: Backtracking with MRV + degree heuristic + LCV + forward checking/arc consistency for hard constraints, followed by min-conflicts or weighted optimization for soft constraints.

- For very large universities, the best practical design is hybrid. First generate a conflict graph and pre-process impossible domain values.
- Then obtain a feasible timetable using propagation-guided backtracking. Finally optimize convenience using soft-constraint penalties.
- When a late change occurs, perform local repair rather than regenerating the complete timetable, thereby preserving stability for students and faculty.

5. STRATEGIC GAME AI FOR REAL-TIME DECISION MAKING

5.1 Minimax Model

- Minimax models the interaction as alternating MAX and MIN levels. MAX represents the AI and chooses the move with the highest utility.
- MIN represents the human opponent and is assumed to choose the move that gives the AI the lowest utility.
- Terminal states contain exact outcomes such as win, loss or draw; when the search stops early, an evaluation function estimates the state value.

Minimax	$V(s) = \max V(\text{child})$ at MAX nodes; $V(s) = \min V(\text{child})$ at MIN nodes. The selected root action maximizes the guaranteed result against an optimal opponent.
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Game component	Minimax representation
Current situation	Root game state
AI legal actions	Branches at MAX node
Opponent responses	Branches at MIN node

Future sequence	Alternating levels of the game tree
Win/loss/resource advantage	Utility or evaluation score
Chosen move	Root child with highest backed-up minimax value

5.2 Computational Challenges

Time complexity	$O(b^m)$ b is branching factor and m is search depth. Even moderate values produce an enormous number of nodes.
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Space complexity	$O(bm)$ with depth-first implementation Memory is lower than breadth-first storage, but repeated state generation is still expensive.
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1. Large branching factor: many units can move, attack, defend or build.
2. Deep tactical consequences: the value of a move may appear several turns later.
3. Real-time limit: the AI cannot wait for a complete tree.
4. Imperfect evaluation: a shallow cutoff may misjudge long-term strategy.
5. Repeated states: different action orders can reach the same position.
6. Uncertainty: hidden information and simultaneous actions may violate pure deterministic minimax assumptions.

5.3 Alpha-Beta Pruning

- Alpha-Beta pruning computes the same optimal move as Minimax for the searched depth but avoids branches that cannot influence the final decision. Alpha (α) is the best value MAX can already guarantee. Beta (β) is the best value MIN can already guarantee. When $\alpha \geq \beta$, further children of that node need not be examined because the opponent already has a better alternative elsewhere.

Pruning rule	Prune when $\alpha \geq \beta$. The remaining branch cannot change the decision at the root.
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Situation	Reasoning
MAX already has value 8	Any alternative below 8 is irrelevant to MAX.
At a MIN node, first child gives 5	MIN can force at most 5 on this branch.
Since $5 \leq \text{MAX}$ guarantee 8	MAX will never choose this branch; remaining MIN children are pruned.

- Move ordering is crucial. If the strongest moves are evaluated first, Alpha-Beta can reduce effective complexity toward $O(b^{(m/2)})$, approximately doubling the searchable depth for the same computation. Poor ordering gives little improvement over plain Minimax.

5.4 Evaluation Function Design

Evaluation	$E(s) = 5M + 4T + 3P + 3R + 2K + 2I - 6D$ M: material/army strength, T: territory control, P: positional advantage, R: resources/economy, K: king/base safety, I: initiative, D: immediate danger.
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Factor	Why it matters
Material / unit strength	More healthy combat units increase tactical power.
Territory / map control	Controls resources, movement routes and strategic locations.
Economy / resources	Enables future production and upgrades.
Base or leader safety	A vulnerable base can make all other advantages irrelevant.

Mobility	More legal and useful moves improve flexibility.
Initiative / tempo	Forcing actions can prevent the opponent from executing a plan.
Threats and tactical patterns	Immediate attacks, traps or capture opportunities require high weight.
Game phase	Weights should change between opening, middle game and endgame.

- Evaluation innovation: Use phase-dependent and normalized features. A resource advantage is important early, while base safety and direct win threats dominate late. This avoids using one rigid scoring rule for every situation.

5.5 Balancing Optimality and Real-Time Performance

- Recommended strategy: Iterative deepening Alpha-Beta with strong move ordering, transposition tables, time control and a domain-specific evaluation function.
 1. Search depth 1, then 2, then 3 and so on until the time budget is nearly exhausted.
 2. Always retain the best move from the last fully completed depth.
 3. Order promising moves first using tactical captures, threats and the previous iteration result.
 4. Store previously evaluated states in a transposition table.
 5. Use quiescence search for unstable tactical states to reduce the horizon effect.
 6. Allocate more time to critical decisions and less time to obvious moves.
 7. Use pruning or beam limits only when necessary, while protecting safety-critical responses.
- This design gives “anytime” behaviour: the AI always has a valid move, but its quality improves as more time becomes available.
- Alpha-Beta preserves Minimax correctness at the completed depth, while iterative deepening makes timing predictable.
- In games with hidden information, expectiminimax, Monte Carlo Tree Search or belief-state methods may be added, but for the stated deterministic adversarial scenario, iterative-deepening Alpha-Beta is the strongest justified choice.

Integrated Comparison and Final Synthesis

Scenario	Why the naive method is insufficient	Best justified solution
Drone	Greedy distance ignores accumulated risk and dynamic blockage.	Risk-aware A* / D* Lite with live replanning.
Traffic	Hill Climbing becomes trapped in local optima.	Adaptive Genetic Algorithm with local refinement.
Mars rover	Offline planning assumes a complete and stable map.	Safe model-based online heuristic search.
Exam timetable	Plain backtracking discovers conflicts too late.	CSP heuristics plus propagation and local repair.
Game AI	Plain Minimax cannot search deeply within time.	Iterative-deepening Alpha-Beta with good evaluation.

Key Learning Outcomes Demonstrated

- Algorithm selection must follow the structure of the environment: static vs dynamic, known vs unknown, single-agent vs adversarial and unconstrained vs constraint-driven.
- A heuristic is valuable only when its accuracy, admissibility, computational cost and safety implications are understood.
- Real-time AI commonly uses bounded, incremental or anytime computation rather than waiting for a theoretically complete search.
- Hybrid methods often outperform a single textbook algorithm by combining global exploration, local refinement, propagation and replanning.
- Safety and explainability should be encoded directly through costs, constraints, evaluation factors and fallback policies.

Conclusion: The most intelligent algorithm is not always the one that searches the deepest or fastest. It is the one that uses available knowledge responsibly, adapts when assumptions fail,

satisfies hard constraints, and produces the best defensible decision within the real-world time and safety limits.

Rubric Alignment Checklist

Rubric Criterion	How this submission achieves Level 4
Understanding of Scenario - 20%	Every scenario is reformulated using states, objectives, uncertainty, constraints and operational context.
Identification of Key Issues - 20%	Dynamic changes, local optima, partial observability, combinatorial explosion and adversarial complexity are explicitly identified.
Application of Concepts - 25%	Formulas, algorithm behaviour, heuristic effects, pruning, propagation and hybrid methods are correctly applied.
Decision Making - 20%	Each scenario ends with a specific recommendation supported by safety, scalability, optimality and real-time reasoning.
Clarity of Response - 15%	Structured headings, comparison tables, formulas, flow steps, highlighted insights and concise conclusions are used throughout.